

ALFAA42378

## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 2-甲氧基乙醇镍, 5% w/v 2-甲氧基乙醇溶液  
 Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

**Cat No. :** 42378  
**Molecular Formula** C6 H14 NiO21

**Supplier** Avocado Research Chemicals Ltd.  
 (Part of Thermo Fisher Scientific)  
 Shore Road, Heysham  
 Lancashire, LA3 2XY,  
 United Kingdom  
 Office Tel: +44 (0) 1524 850506  
 Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
No information available

**Odor**  
Characteristic

#### Emergency Overview

Flammable liquid and vapor. May damage fertility or the unborn child. Harmful if swallowed. Harmful in contact with skin. May cause an allergic skin reaction. Harmful if inhaled. May cause cancer. Causes damage to organs. May cause damage to organs through prolonged or repeated exposure. Toxic to aquatic life with long lasting effects. Moisture sensitive.

#### Classification of the substance or mixture

|                                                      |             |
|------------------------------------------------------|-------------|
| Flammable liquids.                                   | Category 3  |
| Acute Oral Toxicity                                  | Category 4  |
| Acute Dermal Toxicity                                | Category 4  |
| Acute Inhalation Toxicity - Vapors                   | Category 4  |
| Skin Sensitization                                   | Category 1  |
| Carcinogenicity                                      | Category 1B |
| Reproductive Toxicity                                | Category 1B |
| Specific target organ toxicity - (single exposure)   | Category 1  |
| Specific target organ toxicity - (repeated exposure) | Category 2  |
| Chronic aquatic toxicity                             | Category 2  |

#### Label Elements

## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol



## Signal Word

Danger

## Hazard Statements

H226 - Flammable liquid and vapor  
 H317 - May cause an allergic skin reaction  
 H350 - May cause cancer  
 H370 - Causes damage to organs  
 H373 - May cause damage to organs through prolonged or repeated exposure  
 H411 - Toxic to aquatic life with long lasting effects  
 H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled  
 H360 - May damage fertility or the unborn child

## Precautionary Statements

## Prevention

P201 - Obtain special instructions before use  
 P202 - Do not handle until all safety precautions have been read and understood  
 P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
 P233 - Keep container tightly closed  
 P240 - Ground and bond container and receiving equipment  
 P242 - Use non-sparking tools  
 P243 - Take action to prevent static discharges  
 P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
 P264 - Wash face, hands and any exposed skin thoroughly after handling  
 P270 - Do not eat, drink or smoke when using this product  
 P271 - Use only outdoors or in a well-ventilated area  
 P272 - Contaminated work clothing should not be allowed out of the workplace  
 P280 - Wear protective gloves

## Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 P330 - Rinse mouth  
 P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
 P362 + P364 - Take off contaminated clothing and wash it before reuse  
 P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor

## Storage

P403 + P235 - Store in a well-ventilated place. Keep cool

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Vapors may cause flash fire or explosion. Flammable liquid.

## Health Hazards

May damage fertility or the unborn child. Harmful if swallowed. Harmful in contact with skin. May cause an allergic skin reaction. Harmful if inhaled. May cause cancer. Causes damage to organs. May cause damage to organs through prolonged or repeated exposure.

## Environmental hazards

Toxic to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil.

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No | Weight % |
|-----------|--------|----------|
|-----------|--------|----------|

**SAFETY DATA SHEET****Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol**

|                          |             |       |
|--------------------------|-------------|-------|
| 2-Methoxyethanol         | 109-86-4    | 95.00 |
| Nickel 2-methoxyethoxide | 142600-62-2 | 5.00  |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

Difficulty in breathing. May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Storage**

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component        | China                             | Taiwan                                  | Thailand | Hong Kong                               |
|------------------|-----------------------------------|-----------------------------------------|----------|-----------------------------------------|
| 2-Methoxyethanol | TWA: 15 mg/m <sup>3</sup><br>Skin | TWA: 5 ppm<br>TWA: 16 mg/m <sup>3</sup> |          | TWA: 5 ppm<br>TWA: 16 mg/m <sup>3</sup> |

| Component        | ACGIH TLV            | OSHA PEL                                                                                                         | NIOSH                                                       | The United Kingdom                                                                                                 | European Union          |
|------------------|----------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------|
| 2-Methoxyethanol | TWA: 0.1 ppm<br>Skin | (Vacated) TWA: 25 ppm<br>(Vacated) TWA: 80 mg/m <sup>3</sup><br>Skin<br>TWA: 25 ppm<br>TWA: 80 mg/m <sup>3</sup> | IDLH: 200 ppm<br>TWA: 0.1 ppm<br>TWA: 0.3 mg/m <sup>3</sup> | STEL: 3 ppm 15 min<br>STEL: 9 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 3 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 1 ppm (8h)<br>Skin |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography MDHS42/2 Nickel and inorganic compounds of nickel in air (except nickel carbonyl) Laboratory method using flame atomic absorption spectrometry or electrothermal atomic absorption spectrometry

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

## Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Skin and body protection

Long sleeved clothing

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Multi-purpose/ABEK conforming to EN14387 low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

## Appearance

## Physical State

Liquid

## Odor

Characteristic

## Odor Threshold

No data available

## pH

4

## Melting Point/Range

No data available

## Softening Point

No data available

## Boiling Point/Range

No information available

## Flash Point

46 °C / 114.8 °F

Method - No information available

## Evaporation Rate

No data available

## Flammability (solid,gas)

Not applicable

Liquid

## Explosion Limits

No data available

## Vapor Pressure

23 hPa @ 20 °C

## Vapor Density

No data available

(Air = 1.0)

## Specific Gravity / Density

No data available

## Bulk Density

Not applicable

Liquid

## Water Solubility

Immiscible

## Solubility in other solvents

No information available

## Partition Coefficient (n-octanol/water)

## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

|                                  |                          |                                        |
|----------------------------------|--------------------------|----------------------------------------|
| <b>Component</b>                 | <b>log Pow</b>           |                                        |
| 2-Methoxyethanol                 | -0.77                    |                                        |
| <b>Autoignition Temperature</b>  | No data available        |                                        |
| <b>Decomposition Temperature</b> | No data available        |                                        |
| <b>Viscosity</b>                 | No data available        |                                        |
| <b>Explosive Properties</b>      |                          | explosive air/vapour mixtures possible |
| <b>Oxidizing Properties</b>      | No information available |                                        |
| <b>Molecular Formula</b>         | C6 H14 NiO21             |                                        |
| <b>Molecular Weight</b>          | 208.76                   |                                        |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Stability</b>                | Moisture sensitive.                                               |
| <b>Hazardous Reactions</b>      | None under normal processing.                                     |
| <b>Hazardous Polymerization</b> | No information available.                                         |
| <b>Conditions to Avoid</b>      | Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Materials to avoid</b>       | No information available.                                         |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nickel oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

(a) acute toxicity;  
Toxicology data for the components

| Component        | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation             |
|------------------|---------------------------|------------------------------|-----------------------------|
| 2-Methoxyethanol | LD50 = 2370 mg/kg ( Rat ) | LD50 = 1280 mg/kg ( Rabbit ) | LC50 = 1478 ppm ( Rat ) 7 h |

|                                        |                                                           |
|----------------------------------------|-----------------------------------------------------------|
| (b) skin corrosion/irritation;         | No data available                                         |
| (c) serious eye damage/irritation;     | No data available                                         |
| (d) respiratory or skin sensitization; |                                                           |
| Respiratory                            | No data available                                         |
| Skin                                   | Category 1                                                |
|                                        | May cause sensitization by skin contact                   |
| (e) germ cell mutagenicity;            | No data available                                         |
| (f) carcinogenicity;                   | Category 1B                                               |
|                                        | There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;             | Category 1B                                               |
| (h) STOT-single exposure;              | Category 1                                                |

## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

|                                                   |                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Results / Target organs</b>                    | Immune system                                                                                                                                                                                                                                                                 |
| <b>(i) STOT-repeated exposure;</b>                | Category 2                                                                                                                                                                                                                                                                    |
| <b>Target Organs</b>                              | Thymus.                                                                                                                                                                                                                                                                       |
| <b>(j) aspiration hazard;</b>                     | No data available                                                                                                                                                                                                                                                             |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting:<br>Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |

## SECTION 12. ECOLOGICAL INFORMATION

|                            |                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b> | Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system. |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Component        | Freshwater Fish                                                                                                                                                     | Water Flea | Freshwater Algae | Microtox |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|----------|
| 2-Methoxyethanol | LC50: = 9650 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 16000 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 10000 mg/L, 96h static (Lepomis macrochirus) |            |                  |          |

|                                              |                                                                                                                   |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Persistence and Degradability</b>         | Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary |
| <b>Persistence</b>                           | May persist, based on information available.                                                                      |
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.   |

|                                  |                                          |
|----------------------------------|------------------------------------------|
| <b>Bioaccumulative Potential</b> | May have some potential to bioaccumulate |
|----------------------------------|------------------------------------------|

| Component        | log Pow | Bioconcentration factor (BCF) |
|------------------|---------|-------------------------------|
| 2-Methoxyethanol | -0.77   | No data available             |

|                         |                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Mobility in soil</b> | Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility |
|-------------------------|----------------------------------------------------------------------------------------------------------|

|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                   |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition. |

## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

## Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

UN-No UN1188  
Proper Shipping Name ETHYLENE GLYCOL MONOMETHYL ETHER  
Hazard Class 3  
Packing Group III

## IMDG/IMO

UN-No UN1188  
Proper Shipping Name ETHYLENE GLYCOL MONOMETHYL ETHER  
Hazard Class 3  
Packing Group III

## IATA

UN-No UN1188  
Proper Shipping Name ETHYLENE GLYCOL MONOMETHYL ETHER  
Hazard Class 3  
Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component        | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| 2-Methoxyethanol | X                                                   | X                                       | X    | X     | 203-713-7 | X    | X   | X     | X    | X    | X    | KE-23272 |

## National Regulations

| Component                              | Toxic Chemical Substances Control Act |
|----------------------------------------|---------------------------------------|
| 2-Methoxyethanol<br>109-86-4 ( 95.00 ) | Class II (1 wt%)<br>TRQ = 50 kg       |

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Revision Date 09-May-2024



## Nickel 2-methoxyethoxide, 5% w/v in 2-methoxyethanol

## Revision Summary

New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Physical hazards

On basis of test data

## Health Hazards

Calculation method

## Environmental hazards

Calculation method

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**